

Level 2-Programs by-Mr.Kunal

1. Java Program to print the elements of an array

```
public class PrintArray {  
    public static void main(String[] args) {  
        //Initialize array  
        int [] arr = new int [] {1, 2, 3, 4, 5};  
        System.out.println("Elements of given array: ");  
        //Loop through the array by incrementing value of i  
        for (int i = 0; i < arr.length; i++) {  
            System.out.print(arr[i] + " ");  
        }  
    }  
}
```

2. Java Program to copy all elements of one array into another array

```
public class CopyArray {  
  
    public static void main(String[] args) {  
        int []arr1=new int[]{1,2,3,4,5};//1st array  
        int []arr2=new int[arr1.length];//creating 2nd array by array1 ref  
        for(int i=0;i<arr1.length;i++){  
            arr2[i]=arr1[i]; //importing arr1 elements to arr1  
        }  
        //printing 1st array elements  
        System.out.println("1st array elements:");  
        for(int i=0;i<arr1.length;i++){
```

```
System.out.print(arr1[i]+" ");
}
//printing 2nd array elements
System.out.println();
System.out.println("2nd array elements:");
for(int i=0;i<arr2.length;i++){
System.out.print(arr2[i]+" ");
}
}}
```

3. Java Program to print the number of elements present in an array

```
public class NoOfElements {
public static void main(String[] args) {
int arr[]=new int []{1, 2, 3, 4, 2, 7, 8, 8, 3};

int n=arr.length;//stored length in integer n
System.out.println("No of Elements in the array is:"+n);
}

}
```

4.Print duplicate elements

```
import java.util.Scanner;

public class Duplicate {
public static void main(String[] args) {
```

```
int arr[]=new int []{1, 2, 3, 4, 2, 7, 8, 8, 3};  
for(int i=0;i<arr.length;i++){  
for(int j=i+1;j<arr.length;j++){  
if(arr[i]==arr[j]){  
  
System.out.print(arr[j]);  
}}}  
}}
```

5.Assending order

```
import java.util.Scanner;  
public class AssendingOrder {  
  
public static void main(String[] args) {  
Scanner sc = new Scanner(System.in);  
System.out.println("Enter length:");  
int n=sc.nextInt();  
int[]arr=new int[n];  
System.out.println("Enter elements:");  
for(int i=0;i<arr.length;i++){  
arr[i]=sc.nextInt();  
}  
for(int i=0;i<arr.length;i++){  
for(int j=i+1;j<arr.length;j++){  
if(arr[i]>arr[j]){  
int temp;  
temp=arr[i];
```

```

arr[i]=arr[j];
arr[j]=temp;
}
}
}
System.out.println();
System.out.println("Assending order:");
for(int i=0;i<arr.length;i++){
System.out.print(arr[i]+" ");
}
}
}

```

6.Dessending order

```

import java.util.Scanner;

public class DessendingOrder {
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
System.out.println("Enter length:");
int n=sc.nextInt();
int[]arr=new int[n];
System.out.println("Enter elements:");
for(int i=0;i<arr.length;i++){
arr[i]=sc.nextInt();
}
for(int i=0;i<arr.length;i++){
for(int j=i+1;j<arr.length;j++){

```

```

if(arr[i]<arr[j]){
int temp;
temp=arr[i];
arr[i]=arr[j];
arr[j]=temp;
}
}
}
System.out.println();
System.out.println("Dessending order:");
for(int i=0;i<arr.length;i++){
System.out.print(arr[i]+" ");
}
}

}

```

7.Find any smallest number means(1st or 2nd or (any)th) in Array

```

import java.util.Scanner;

public class Pl {

public static int  gSS(int a[],int total){
for(int i=0;i<total;i++){
for(int j=i+1;j<total;j++){
if(a[i]>a[j]){
int temp;
temp=a[i];

```

```

a[i]=a[j];
a[j]=temp;
}
}
}

return a[1]; //guys this is the key operator
//while finding smallest element if you enter a[0]
//means it ll return 1st smallest
//if you enter a[1] means it ll return 2nd largest number
}

public static void main(String[] args) {

Scanner sc = new Scanner(System.in);
System.out.println("Enter length:");
int n=sc.nextInt();
int[] a=new int[n];
System.out.println("Enter elements:");
for(int i=0;i<a.length;i++){
a[i]=sc.nextInt();
}
System.out.println("Second smallest: "+gSS(a,a.length));
}
}

```

8. Find any largest number means(1st or 2nd or (any)th) in Array

```
import java.util.Scanner;

public class Pl {

    public static int    gSS(int a[],int total){
        for(int i=0;i<total;i++){
            for(int j=i+1;j<total;j++){
                if(a[i]>a[j]){
                    int temp;
                    temp=a[i];
                    a[i]=a[j];
                    a[j]=temp;
                }
            }
        }

        return a[total-2]; //guys this is the key operator
        //while finding smallest element if you enter a[total-1]
        //means it ll return 1st largest
        //if you enter a[total-2] means it ll return 2nd largest number
    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        System.out.println("Enter length:");
        int n=sc.nextInt();
        int[]a=new int[n];
        System.out.println("Enter elements:");
        for(int i=0;i<a.length;i++){
            a[i]=sc.nextInt();
        }
    }
}
```

```
}
```

```
System.out.println("Second greatest: "+gSS(a,a.length));
```

```
}
```

```
}
```

9. Java Program to find the frequency of each element in the array

```
public class Frequency {  
  
    public static void main(String[] args) {  
        int []arr=new int[]{1, 2, 8, 3, 2, 2, 2, 5, 1};  
        int []fr=new int[arr.length];  
  
        int visited =-1;  
        for(int i=0;i<arr.length;i++){  
            int count=1;  
            for(int j=i+1;j<arr.length;j++){  
                if(arr[i]==arr[j]){  
                    count++;  
                    fr[j]=visited;  
                }  
            }  
            if(fr[i]!=visited)  
                fr[i]=count;  
        }  
        System.out.println(" .....");  
    }  
}
```



```

System.out.println("Elements||Frequency");
System.out.println(".....");
for(int i=0;i<fr.length;i++){
if(fr[i]!=visited)
System.out.println("    " + arr[i] + "    |    ");
}
System.out.println(".....");
}
}

```

10. Java Program to left rotate the elements of an array

```

public class LeftRotate {

    public static void main(String[] args) {
        int []arr=new int[]{1, 2, 3,4,5};
        int n=3;
        System.out.println("Original array:");
        for(int i=0;i<arr.length;i++){
            System.out.print(arr[i]+" ");
        }
        for(int i=0;i<n;i++){
            int j,first;
            first=arr[0];
            for(j=0;j<arr.length-1;j++){
                arr[j]=arr[j+1];
            }
            arr[j]=first;
        }
    }
}

```

```

}
System.out.println();
System.out.println(" Array after left rotation: ");
for(int i1 = 0; i1< arr.length; i1++){
System.out.print(arr[i1] + " ");
}

}

}

```

10b. Java Program to right rotate the elements of an array

```

public class RightRotate {

    public static void main(String[] args) {
        int []arr=new int[]{1, 2, 3,4,5};
        int n=3;
        System.out.println("Original array:");
        for(int i=0;i<arr.length;i++){
            System.out.print(arr[i]+" ");
        }
        for(int i=0;i<n;i++){
            int j,last;
            last=arr[arr.length-1];
            for(j=arr.length-1;j>0;j--){
                arr[j]=arr[j-1];
            }
            arr[0]=last;
        }
    }
}

```

```
System.out.println();  
System.out.println(" Array after right rotation: ");  
for(int i1 = 0; i1< arr.length; i1++){  
System.out.print(arr[i1] + " ");  
}  
}  
}
```